

USA State Socioeconomic and Crime Data Analysis (2013–2022)

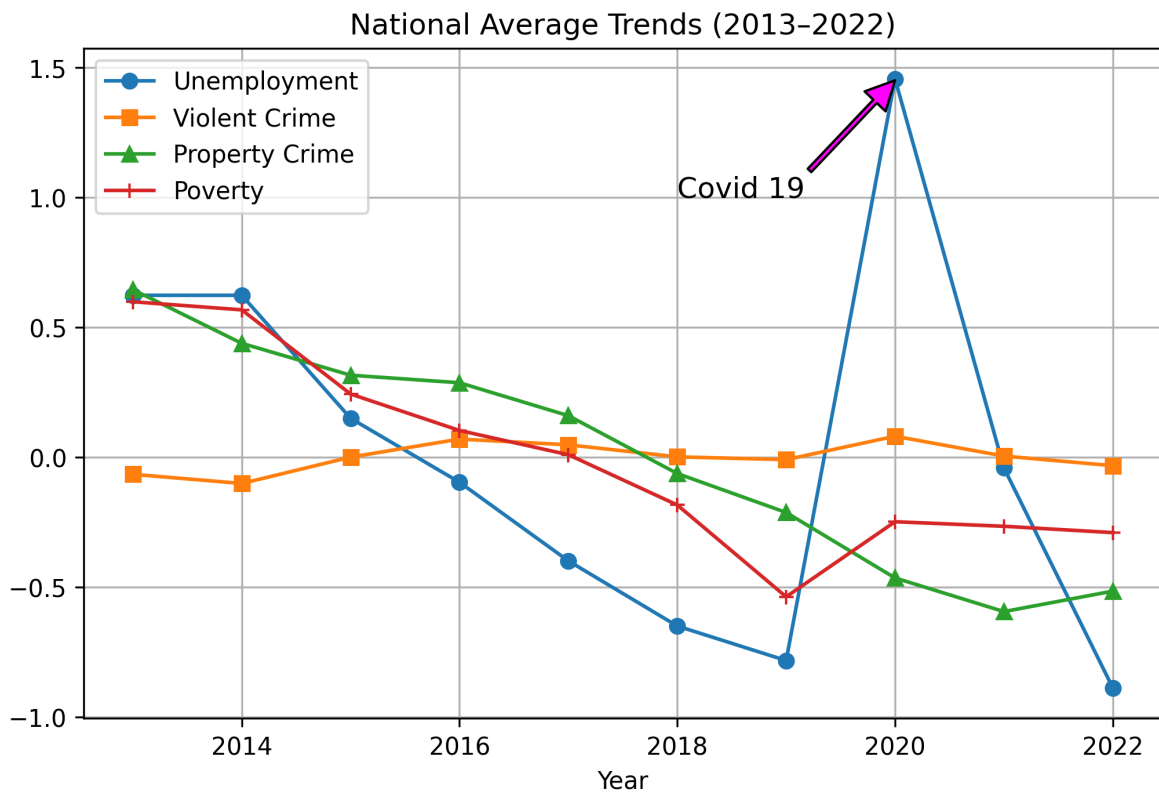
Data Analysis Project

Mehraveh Goharshadi

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National Average Trends report



Between 2013 and 2019, the United States saw steady improvements across most socioeconomic and public-safety indicators: unemployment fell, poverty rates declined, and both violent and property crime went downward. The COVID-19 pandemic in 2020, however, produced a sharp—but temporary—disruption, most notably in the labor market.

Unemployment Rate vs. Violent Crime Rate Report

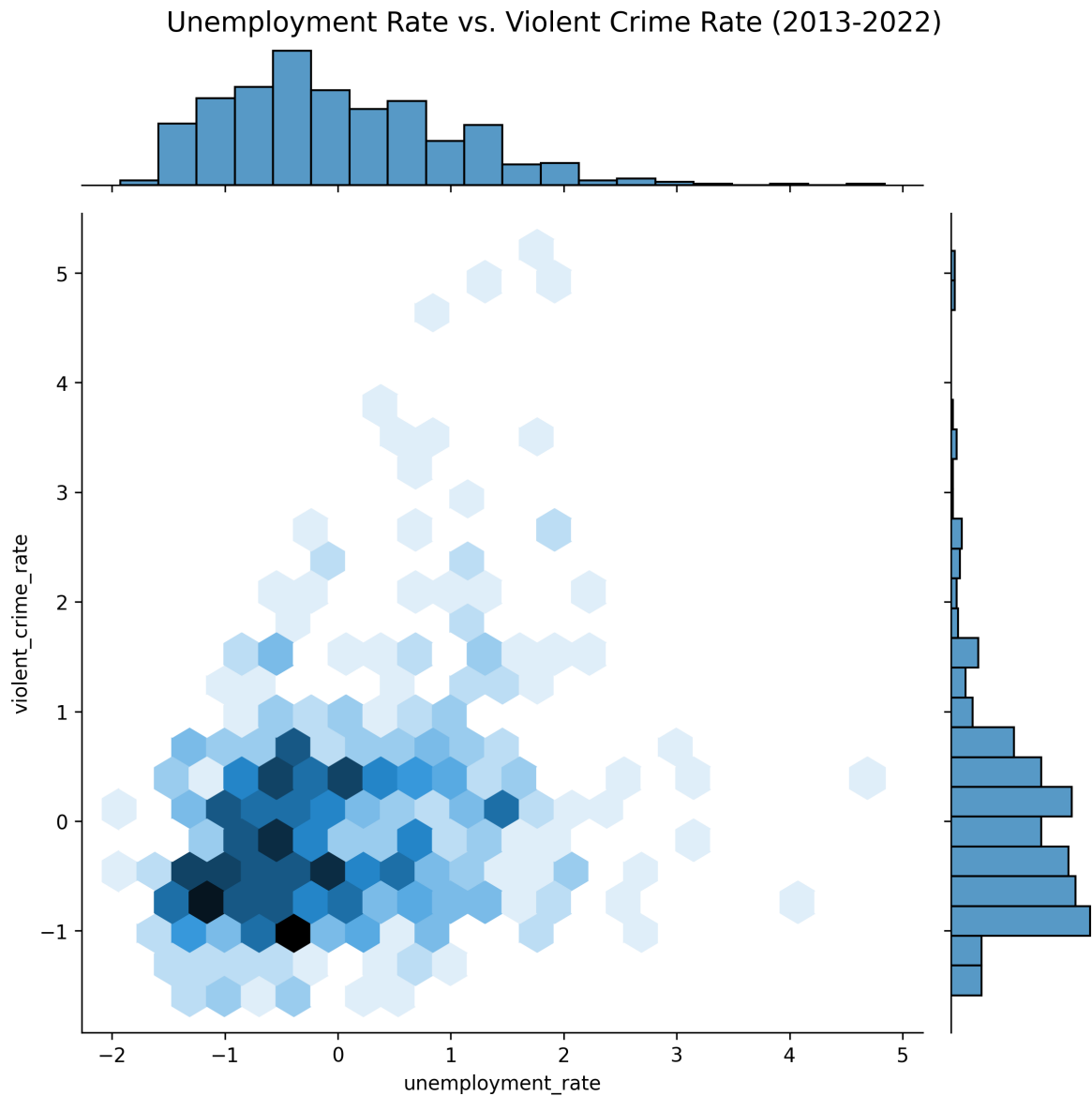


Figure 1: $p_coeff: 0.30$, $p_value: 0.00$

While the correlation between unemployment and violent crime is weak (only 0.30), the fact that the p-value is very low indicates that this relationship is statistically significant. However, the strength of the correlation is not strong enough to conclude that unemployment has a major influence on violent crime.

Poverty Rate vs. White Population Rate Report

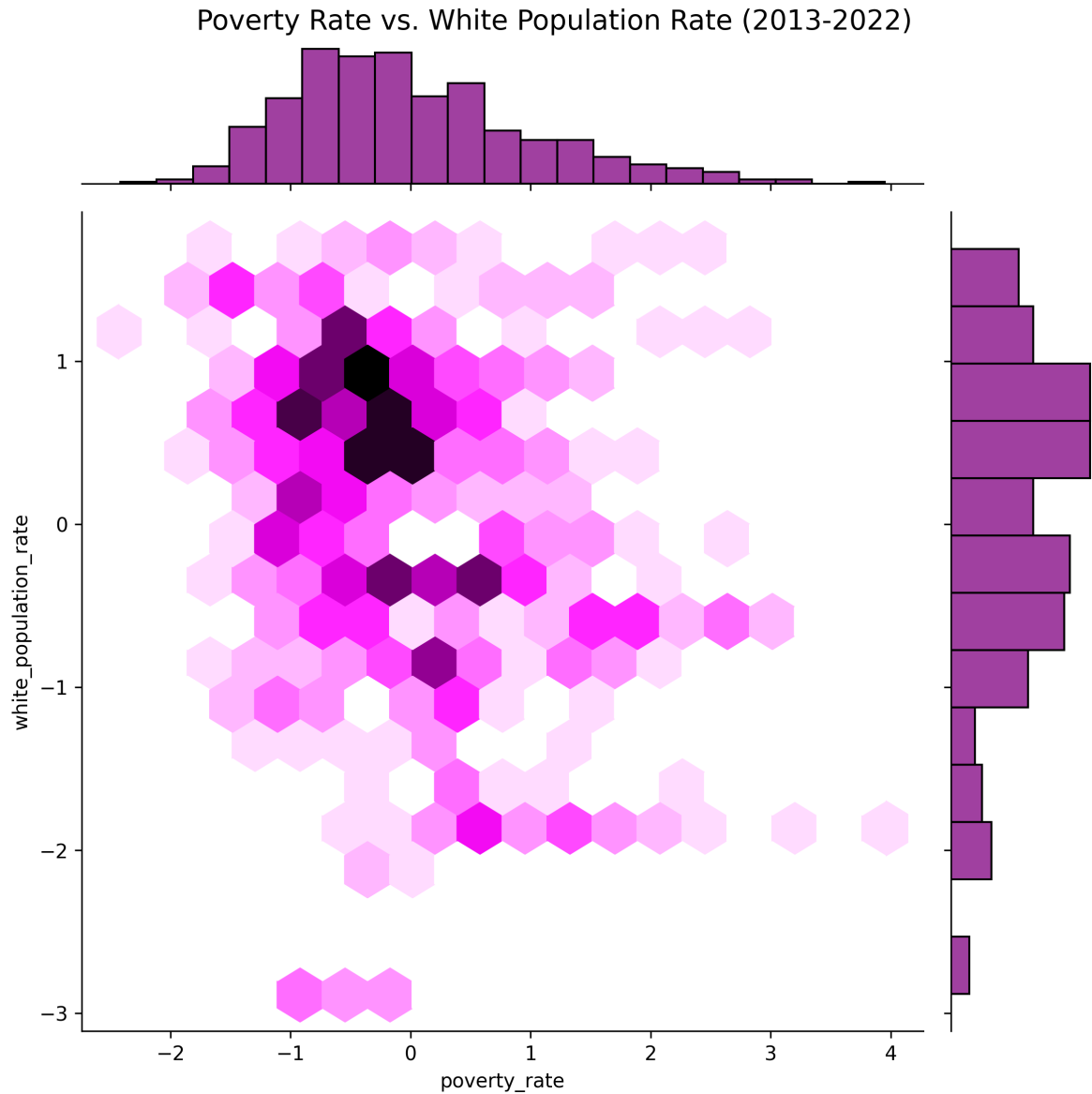
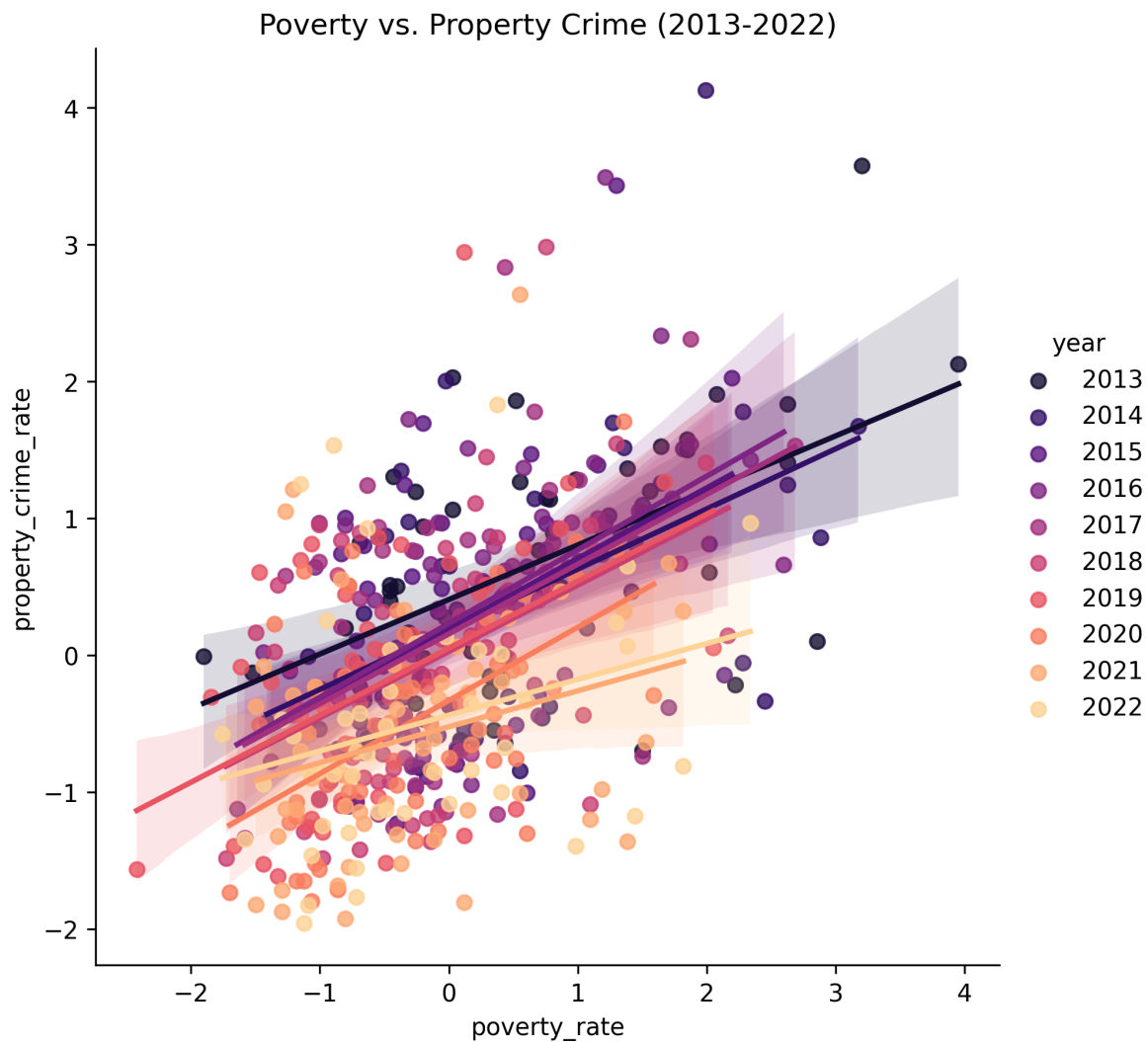


Figure 2: p_coeff : -0.26, p -value: 0.00

While the correlation between these two is weak (with a value of -0.26), the statistical significance (as indicated by the very low p -value) means we can be confident that there is a genuine, though weak, relationship between these two variables. In other words: Where the proportion of the white population is higher, the poverty rate tends to be slightly

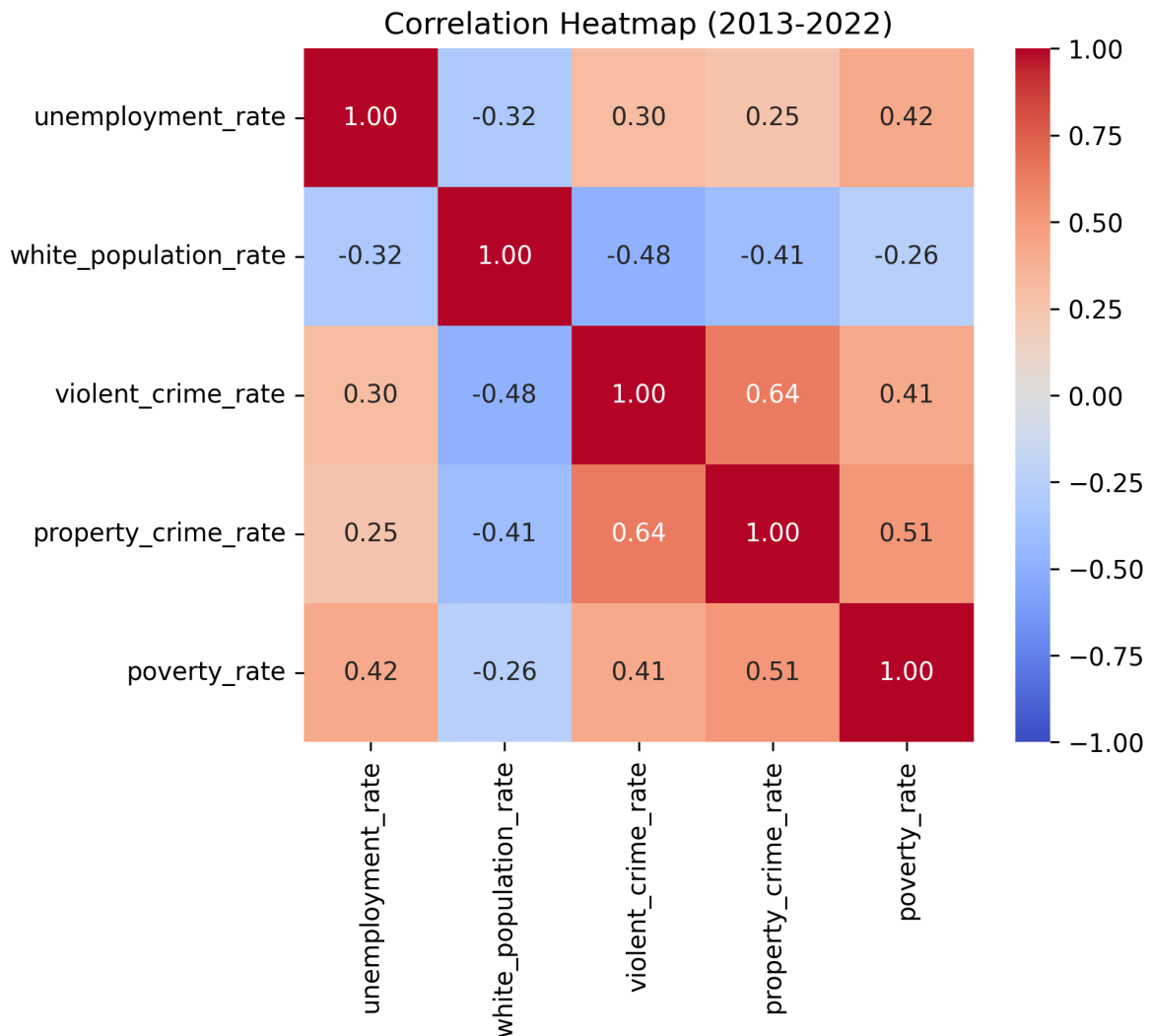
lower.

Poverty vs. Property Crime Report



This plot clearly indicates that there is a positive relationship between poverty rates and property crime rates from 2013 to 2022. Over the years, this relationship has remained mostly consistent. It supports the idea that states or years with higher poverty levels tend to experience higher property crime rates.

Correlation Heatmap (2013-2022) Report

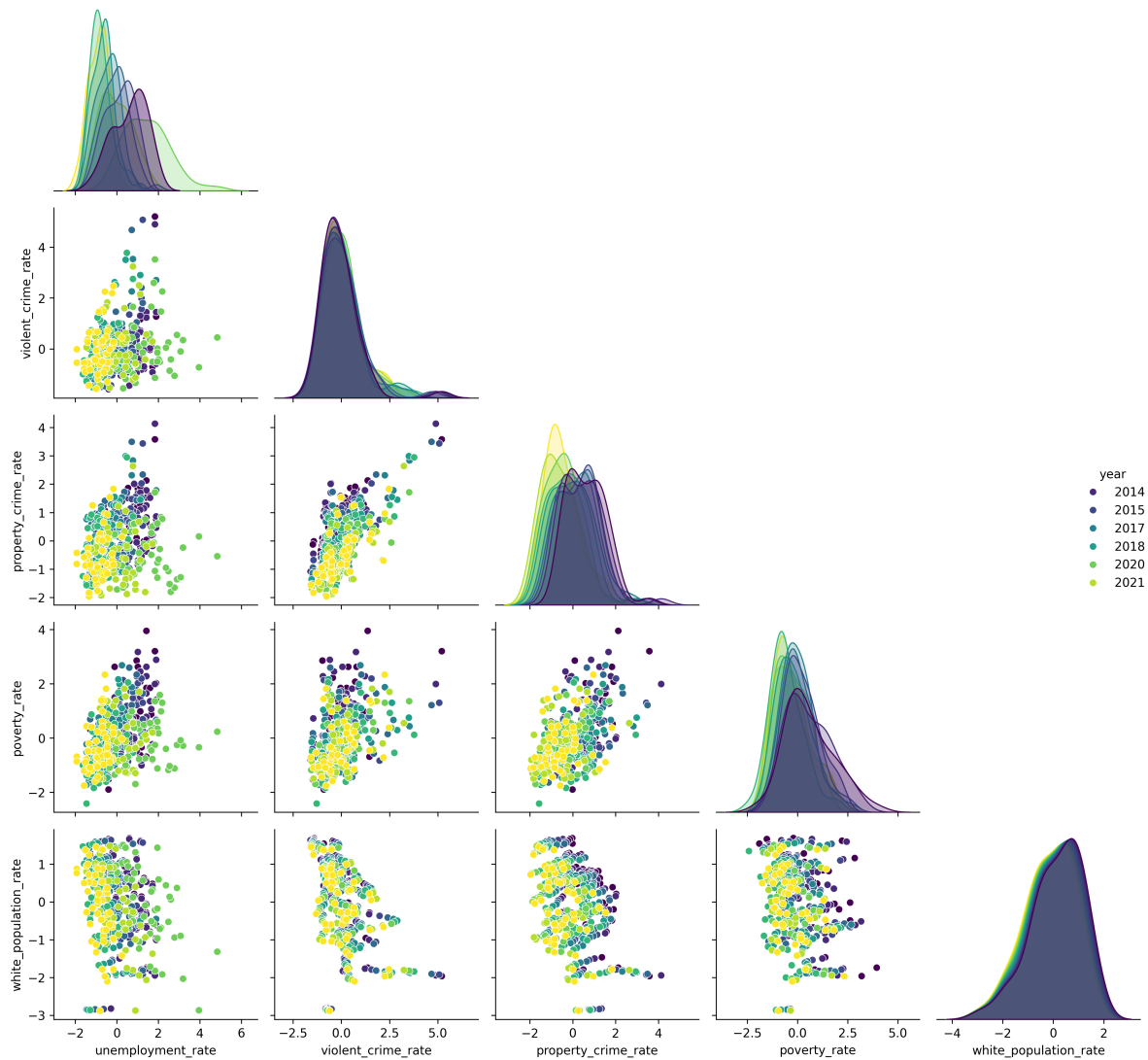


1. Poverty and Crime: There is a moderate positive relationship between poverty rates and both property crime and violent crime, indicating that as poverty increases, both types of crime tend to rise as well.

2. Areas with higher white population rates tend to have lower crime rates, both for violent and property crimes

3. There is a strong positive correlation between property crime and violent crime, meaning areas with higher property crime tend to have higher violent crime as well.

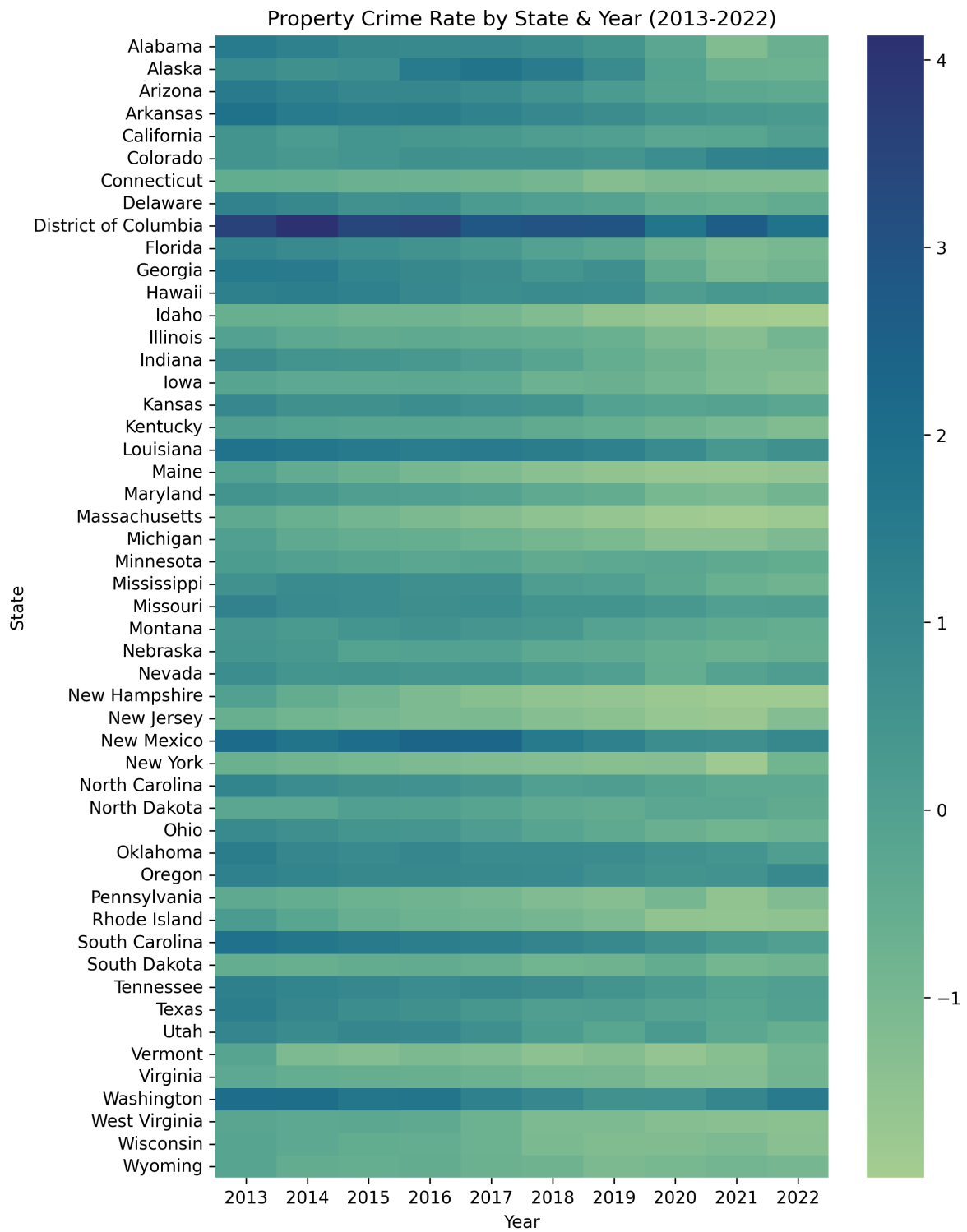
Pairplot (2013-2022) Report



2014-2018: These years show relatively stable distributions for most variables, with only modest changes.

2020-2021: The most significant changes happen here, with poverty and crime rates showing increased values, which likely corresponds to the economic difficulties during the pandemic.

Property Crime Rate by State & Year Report



This heatmap reveals how property crime rates vary by state and year, showing a long-term decline in most states.

Top 5 highest average property-crime states:

1.District of Columbia

2.New Mexico

3.Washington

4.Louisiana

5.South Carolina

Top 5 lowest average property-crime states:

1.New York

2.New Jersey

3.Idaho

4.New Hampshire

5.Massachusetts

Top 5 highest average violent-crime states:

1.District of Columbia

2.Alaska

3.New Mexico

4.Tennessee

5.Arkansas

Top 5 lowest average violent-crime states:

1.Virginia

2.Connecticut

3.New Hampshire

4.Vermont

5.Maine

Property Crime Rate (2013-2017 vs 2018-2022) Report

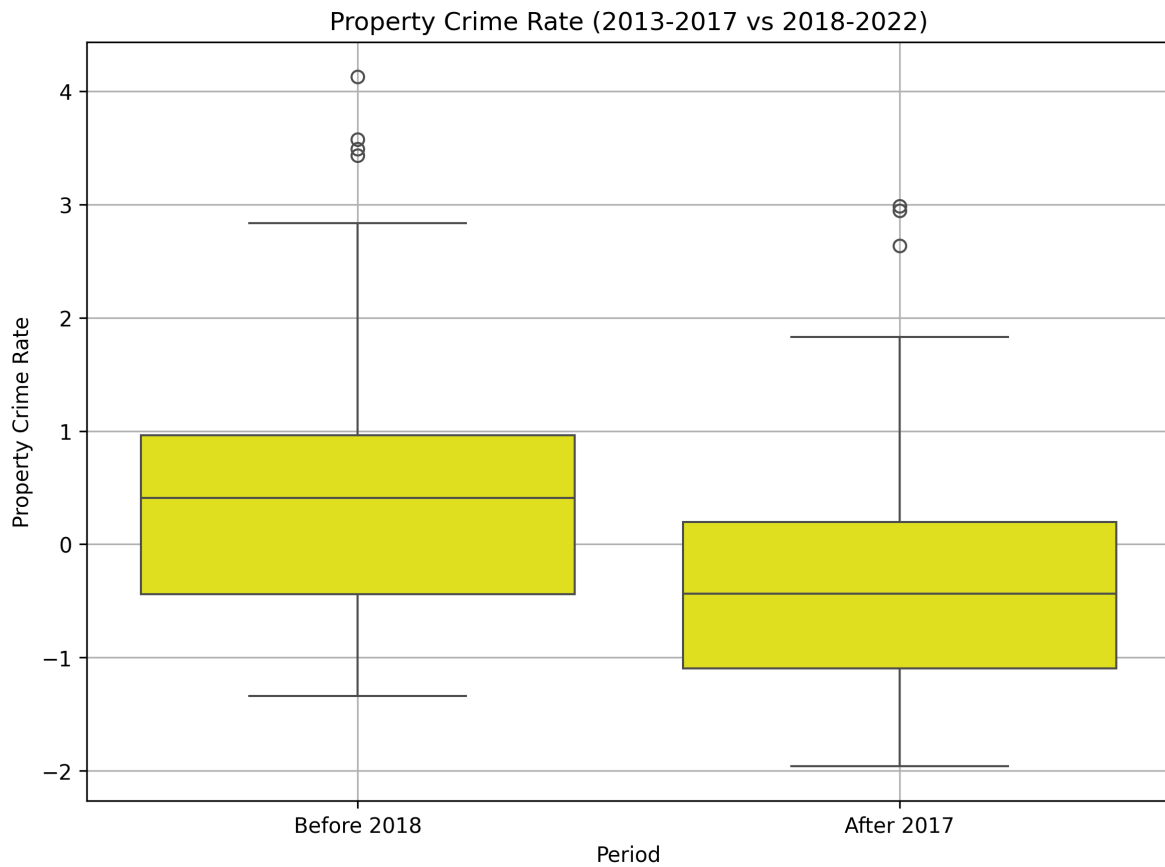


Figure 3: T-stat: 8.96, P-value: 0.00

The T-test results show that there is a statistically significant difference between the property crime rates before 2018 and after 2017.

The boxplot indicates that property crime rates were generally higher and more variable before 2018. After 2017, property crime rates decreased overall.

Viloent Crime Rate (2013-2017 vs 2018-2022) Report

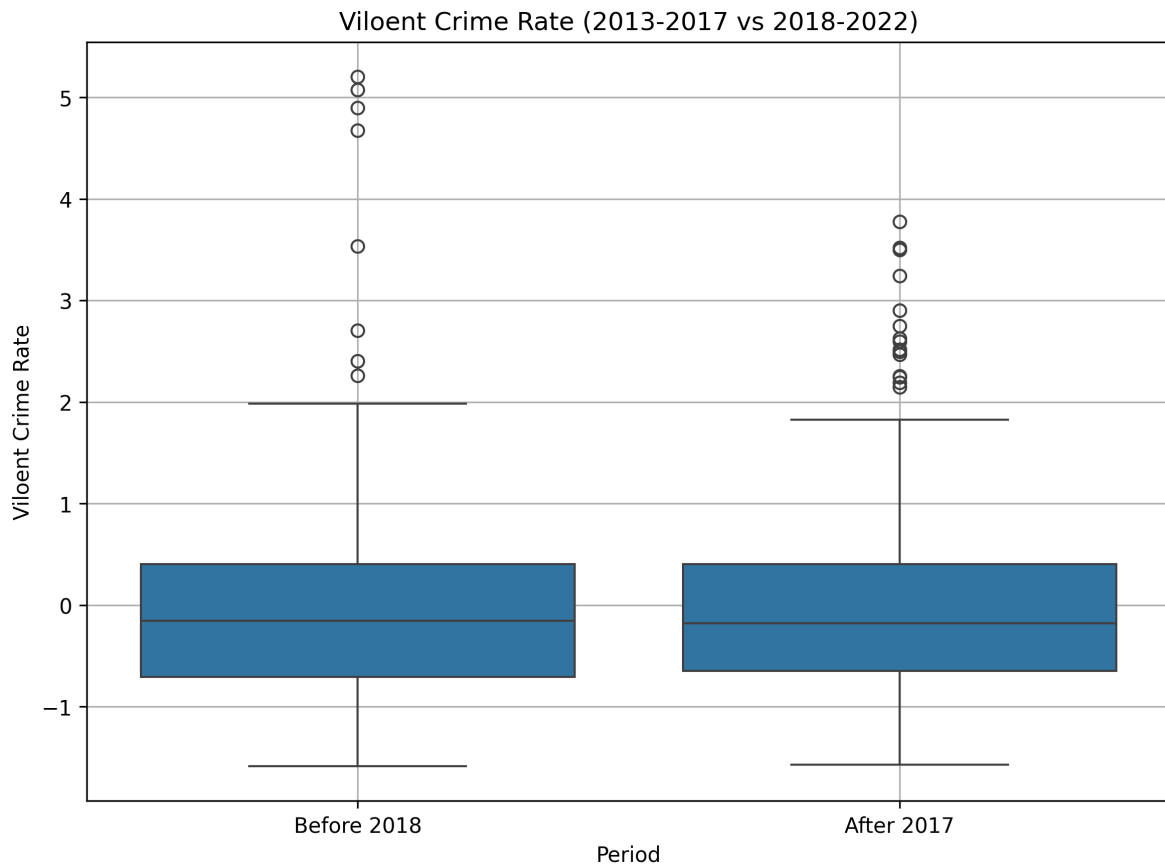
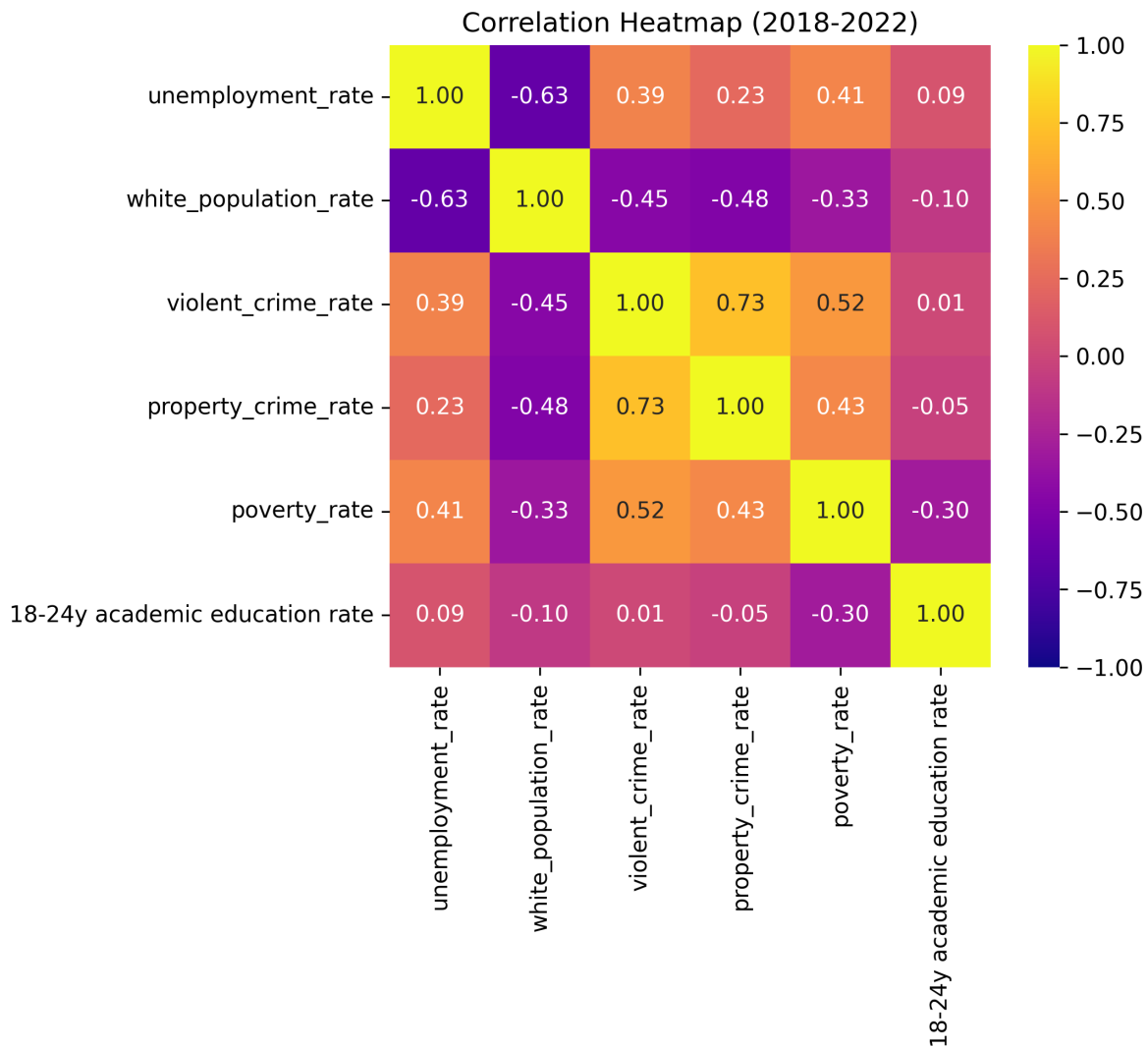


Figure 4: T-stat: -0.21, P-value: 0.83

There is no significant difference between violent crime rates before 2018 and after 2017. Also the T-statistic close to 0 and the high p-value (0.83) suggest that violent crime rates have remained relatively consistent across these two periods.

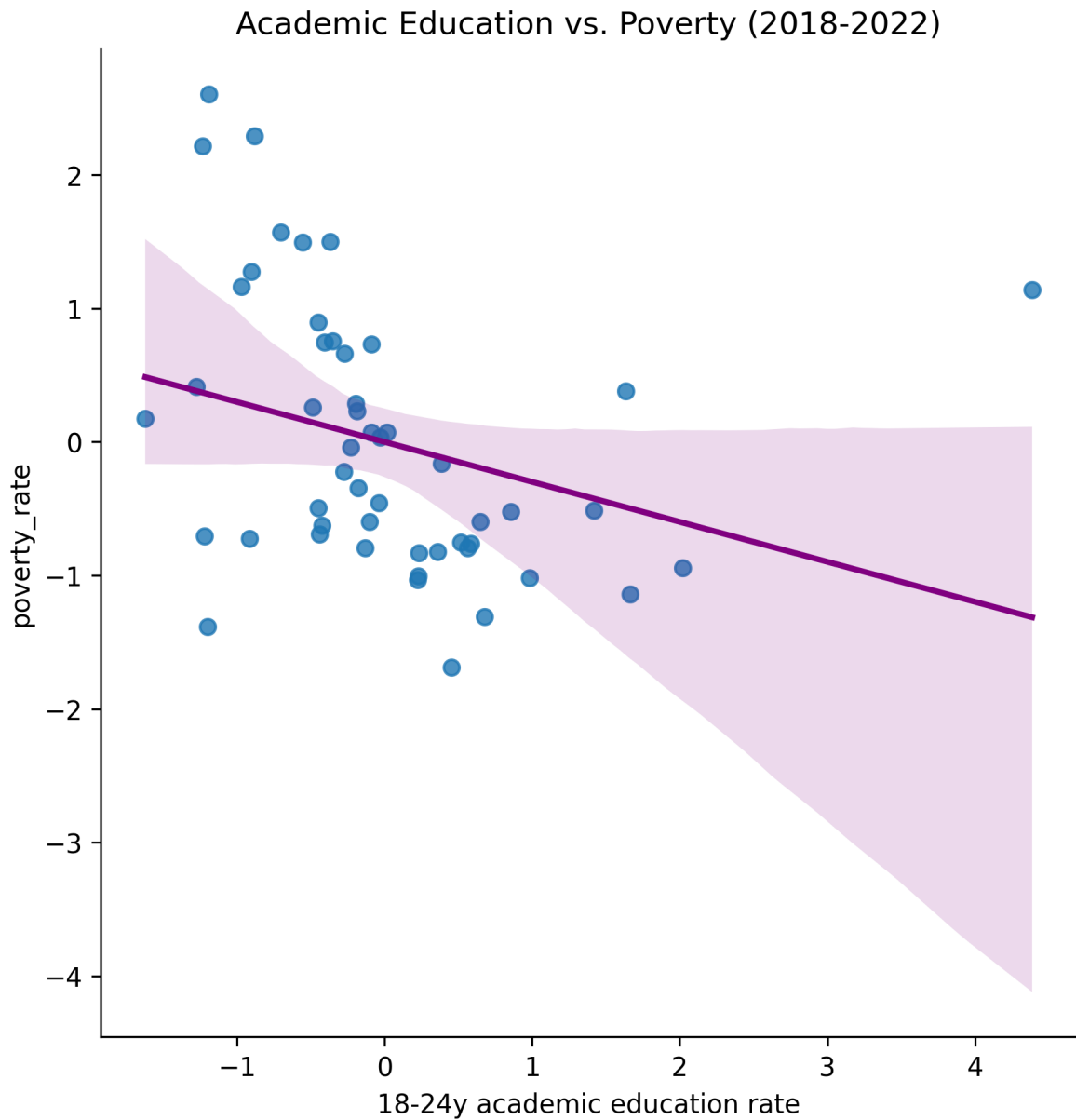
Correlation Heatmap (2018-2022) Report



Academic education rates for 18-24-year-olds have weak correlations with the other variables, but they show a slight negative relationship with poverty, which might suggest that higher education levels could have some effect on reducing poverty.

Since poverty had a moderate positive correlation with crime rates, perhaps Academic education has an indirect negative relation with crime rates which is rather weak.

Academic Education vs. Poverty (2018-2022) Report



States with higher levels of academic education in this age group tend to have lower poverty rates, which supports the idea that education can play a role in reducing poverty. However, the relationship is not perfectly linear, and there are exceptions where states

with high education levels still experience relatively high poverty, suggesting that other factors also influence poverty rates.